- The lag order can be chosen by the same criteria (such as AIC, BIC, and sequential likelihood ratio tests) for the VAR model in levels before the cointegrating rank is tested.

3.5.4 Maximum Likelihood Estimation of the cointegrated VECM

- For simplicity, consider the model without deterministic terms,

$$\Delta \mathbf{y}_t = -\mathbf{\Pi}\mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t. \tag{21}$$

- If it is found that $0 < \text{Rank}(\mathbf{\Pi}) = m < k$, then the matrix $\mathbf{\Pi}$ can be represented as $\mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'$, where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $k \times m$ matrices with full column ranks.

- The cointegrated VECM (21) becomes a *reduced rank multivariate regression*. Johansen derived the MLE of the parameters under the rank restriction $\text{Rank}(\mathbf{\Pi}) = m$ and shows that $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{v}}_1, \hat{\boldsymbol{v}}_2, \ldots, \hat{\boldsymbol{v}}_m)$, where $\hat{\boldsymbol{v}}_i$ are the eigenvectors associated with the eigenvalues $\hat{\lambda}_i$, where $\hat{\lambda}_1 > \hat{\lambda}_2 > \ldots > \hat{\lambda}_k$.

- Recall that the factorization $\mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'$ is not unique, but in the ML estimation, we do not assume that $\boldsymbol{\beta}$ is normalized. From the estimate $\hat{\boldsymbol{\beta}}$, one can form the normalized estimator $\hat{\boldsymbol{\beta}}_c$ by imposing appropriate normalization and identifying constraints, where c signifies constraints.

- For instance, one could set $\boldsymbol{\beta} = (\mathbf{I}_m, \boldsymbol{\beta}_1')'$, where $\boldsymbol{\beta}_1$ is a $(k-m) \times m$ matrix. The MLE of $\boldsymbol{\beta}_1$ can be obtained by the last $k-m$ rows of $\hat{\boldsymbol{\beta}}\hat{\boldsymbol{\beta}}_{(m)}^{-1}$, where $\hat{\boldsymbol{\beta}}_{(m)}$ denotes the first m rows of $\hat{\boldsymbol{\beta}}$.

- After obtaining $\hat{\boldsymbol{\beta}}_c$, the MLE of the remaining parameters can be obtained by the multivariate OLS regression

$$\Delta \mathbf{y}_t = -\boldsymbol{\alpha}\left(\hat{\boldsymbol{\beta}}_c' \mathbf{y}_{t-1}\right) + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t. \tag{22}$$

  That is, one can estimate the short-run parameters $(\boldsymbol{\alpha}, \mathbf{\Gamma}_1, \ldots, \mathbf{\Gamma}_p)$ by regressing $\Delta \mathbf{y}_t$ on $\left(\hat{\boldsymbol{\beta}}_c' \mathbf{y}_{t-1}\right)$, $\Delta \mathbf{y}_{t-1}$, …, and $\Delta \mathbf{y}_{t-p+1}$.

- The normalization from $\hat{\boldsymbol{\beta}}$ to $\hat{\boldsymbol{\beta}}_c$ only affects the estimates of $\boldsymbol{\alpha}$ but not the estimates of the other parameters in the VECM.
- $\hat{\boldsymbol{\beta}}_c$ is T-consistent (super-consistent), while the ML estimators of the short-run parameters are $\sqrt{T}$-consistent.
- It should be stressed that, having established the long-run relations, all the RHS regressors in (22) are stationary, and the standard OLS estimation and testing methods are applicable. The estimators of the short-run parameters have the same asymptotic properties we would obtain if we knew the true $\boldsymbol{\beta}$.
- If no restrictions are imposed on $(\boldsymbol{\alpha}, \boldsymbol{\Gamma}_1, \ldots, \boldsymbol{\Gamma}_p)$, the RHS variables are the same across all the equations in the VECM, and OLS estimation can be used without loss of asymptotic efficiency.
- Deterministic terms can be easily included in the model without additional computational difficulty. The asymptotic properties of the parameter estimators are essentially unchanged, although the convergence rates for the deterministic terms depend on the specific terms included and may be different from $\sqrt{T}$.

4 Structural VARs

- The reduced-form VAR

$$\mathbf{y}_t = \boldsymbol{\phi}_0 + \boldsymbol{\Phi}_1 \mathbf{y}_{t-1} + \boldsymbol{\Phi}_2 \mathbf{y}_{t-2} + \ldots + \boldsymbol{\Phi}_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t$$
$$E\left(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'\right) = \boldsymbol{\Sigma}$$

  has two main shortcomings:

  - It does not explicitly allow for contemporaneous relations among its variables.
  - The reduced-form errors, $\boldsymbol{\varepsilon}_t$, are generally correlated, but we wish to answer, "What is the effect of a shock to one variable, holding all other shocks constant?" The Cholesky decomposition approach is arbitrary and lacks economic content.

- These issues motivate the *structural VAR (SVAR)* model

$$\mathbf{A}\mathbf{y}_t = \mathbf{A}_1^*\mathbf{y}_{t-1} + \ldots + \mathbf{A}_p^*\mathbf{y}_{t-p} + \mathbf{B}\mathbf{u}_t, \tag{23}$$

  where $\mathbf{A}_i^*$, for $i = 1, \ldots, p$, are $k \times k$ structural parameter matrices, which generally differ from their reduced form counterparts, and the structural errors, $\mathbf{u}_t$, are white noise with a diagonal covariance matrix $\mathbf{\Omega} \equiv E(\mathbf{u}_t\mathbf{u}_t')$.

- The goal is to estimate $\mathbf{A}$, $\mathbf{B}$, $\mathbf{A}_i^*$, and $\mathbf{\Omega}$. However, the model's parameters are not identified at this level of generality.

- The SVAR is an example of the simultaneous equation models (SEMs). One cannot estimate the SVAR by OLS equation-by-equation due to simultaneity bias.

- We can always estimate the reduced-form matrices $\mathbf{\Phi}_i$ and $\mathbf{\Sigma}$ from the VAR model. Then how to recover (or identify) $(\mathbf{A}, \mathbf{B}, \mathbf{A}_i^*, \mathbf{\Omega})$ using the information on $(\mathbf{\Phi}_i, \mathbf{\Sigma})$?

- Assuming that $\mathbf{A}$ is nonsingular, then

$$\mathbf{y}_t = \mathbf{A}^{-1}\mathbf{A}_1^*\mathbf{y}_{t-1} + \ldots + \mathbf{A}^{-1}\mathbf{A}_p^*\mathbf{y}_{t-p} + \mathbf{A}^{-1}\mathbf{B}\mathbf{u}_t,$$

  which implies

$$\mathbf{A}^{-1}\mathbf{A}_i^* = \mathbf{\Phi}_i, \quad i = 1, 2, \ldots, p,$$
$$\mathbf{A}^{-1}\mathbf{B}\mathbf{\Omega}\mathbf{B}'\mathbf{A}^{-1\prime} = \mathbf{\Sigma}, \quad (\text{from } \boldsymbol{\varepsilon}_t = \mathbf{A}^{-1}\mathbf{B}\mathbf{u}_t).$$

- Since the covariance matrix, $\mathbf{\Sigma}$, must be symmetric, we have $k(k+1)/2$ independent equations. However, $\mathbf{A}$ and $\mathbf{B}$ are $k \times k$ matrices, and $\mathbf{\Omega}$ has k nonzero elements. To obtain unique $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{\Omega}$ matrices from $\mathbf{\Sigma}$, we need additional $2k^2 + k - \frac{1}{2}k(k+1)$ *a priori* identifying restrictions that are deduced from economic reasoning.

- Three types of SVAR models have been proposed

– **A**-model: $\mathbf{B} = \mathbf{I}_k$. We need (at least) $k^2 + k - \frac{1}{2}k\,(k+1) = \frac{1}{2}k\,(k+1)$ additional restrictions placed on $\mathbf{A}$ and $\boldsymbol{\Omega}$.

– **B**-model: $\mathbf{A} = \mathbf{I}_k$. If the variances of the structural errors are normalized to one (i.e., $\boldsymbol{\Omega} = \mathbf{I}_k$) or restricted to certain theoretical values, then we need $\frac{1}{2}k\,(k-1)$ additional restrictions placed on $\mathbf{B}$.

– **AB**-model: If $\boldsymbol{\Omega} = \mathbf{I}_k$, we need $2k^2 - \frac{1}{2}k\,(k+1)$ restrictions, which can be placed on both $\mathbf{A}$ and $\mathbf{B}$.

4.1 The A-Model

- In this case, $\mathbf{B} = \mathbf{I}_k$, and equation (23) reduces to

$$\mathbf{A}\mathbf{y}_t = \mathbf{A}_1^*\mathbf{y}_{t-1} + \ldots + \mathbf{A}_p^*\mathbf{y}_{t-p} + \mathbf{u}_t, \tag{24}$$

  where $\mathbf{u}_t = \mathbf{A}\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\Omega} = E\,(\mathbf{u}_t\mathbf{u}_t')$ is a diagonal matrix.

- The MA representation of (24) based on $\mathbf{u}_t$ is given by

$$\mathbf{y}_t = \boldsymbol{\Theta}_0\mathbf{u}_t + \boldsymbol{\Theta}_1\mathbf{u}_{t-1} + \boldsymbol{\Theta}_2\mathbf{u}_{t-2} + \ldots, \tag{25}$$

  where $\boldsymbol{\Theta}_i = \boldsymbol{\Psi}_i\mathbf{A}^{-1}$, for $i = 0, 1, 2, \ldots$, with $\boldsymbol{\Psi}_i = \sum_{j=1}^{i}\boldsymbol{\Phi}_j\boldsymbol{\Psi}_{i-j}$ (see Section 2.3). If the structural model (24) is identified, the corresponding impulse responses will be unique.

- A conventional identifying strategy for the **A**-model is to assume that $\mathbf{A}$ is a lower-triangular matrix with a unit main diagonal,

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & \ldots & 0 \\ a_{21} & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ a_{k1} & a_{k2} & & 1 \end{pmatrix}.$$

  This imposes a total of $k\,(k+1)\,/2$ restrictions – k normalization restrictions and $k\,(k-1)\,/2$ exclusion restrictions. Under these restrictions, the SVAR model is just-identified.

- The lower-triangular form of $\mathbf{A}$ implies that y_{jt} has no contemporaneous effect on $y_{\ell t}$ in the preceding equations ($\ell < j$). We can treat y_{jt} in the jth equation as the LHS variable by the normalization restrictions.

- If $\mathbf{A}$ is lower-triangular, the same is true for $\mathbf{A}^{-1}$. Since $\mathbf{\Sigma} = \mathbf{A}^{-1}\mathbf{\Omega}\mathbf{A}^{-1\prime}$ and $\mathbf{\Omega}$ is a diagonal matrix, applying the LDL decomposition to $\mathbf{\Sigma}$ gives us $\mathbf{A}$ and $\mathbf{\Omega}$.

- Compared to the Cholesky decomposition of $\mathbf{\Sigma}$, the diagonal elements of L are required to be one at the cost of introducing an additional diagonal matrix in the decomposition. Therefore, the impulse responses under these restrictions are qualitatively the same as the orthogonalized impulse responses based on the Cholesky decomposition of $\mathbf{\Sigma}$ we considered earlier, except that $\mathbf{v}_t$ have unit variances but $\mathbf{u}_t$ may not.

- Other restrictions (not necessarily exclusion restrictions) can be placed on $\mathbf{A}$ that do not yield a lower triangular matrix. See, for example, Blanchard and Watson (1986), and Bernanke (1986), and Sims (1986).

- It should be noted that the identifying restrictions for $\mathbf{A}$ cannot be arbitrary restrictions. See Lütkepohl (2005, Proposition 9.1) for a necessary and sufficient *rank condition* for local identification of the $\mathbf{A}$-model.

4.2 The B-Model

- In this case, $\mathbf{A} = \mathbf{I}_k$ and $\boldsymbol{\varepsilon}_t = \mathbf{B}\mathbf{u}_t$. If we normalize $\mathbf{\Omega} = \mathbf{I}_k$, then $\mathbf{\Sigma} = \mathbf{B}\mathbf{B}'$. We need $k\left(k-1\right)/2$ additional restrictions on $\mathbf{B}$ for exact identification.

- This model identifies the structural shocks from the reduced-form residuals only and does not place restrictions on the contemporaneous relationships between the component series directly.

- For example, we again can choose $\mathbf{B}$ to be lower-triangular, which provides the minimum number of restrictions for identification. Note, however, that this recursive structure is chosen based on some theoretical justifications so that $\mathbf{u}_t$ can be viewed as structural shocks.

- The triangular form is just one example; one could also place coefficient restrictions such that one variable has no instantaneous effect on another without relying on a triangular formulation.

- Other types of restrictions on $\mathbf{B}$ can be imposed, such as symmetry restrictions (Enders and Souki, 2008) and sign restrictions (Mountford and Uhlig, 2009; Uhlig, 2017).

- See Lütkepohl (2005, Proposition 9.2) for a necessary and sufficient rank condition for local identification of the $\mathbf{B}$-model.

4.3 Identification by Long-Run Restrictions

- Imposing restrictions directly on $\mathbf{A}$ and $\mathbf{B}$ matrices is not necessary for identification. Another type of restrictions was proposed by Blanchard and Quah (1989), who suggest identifying the structural model by assuming that some shocks do not have long-run effects.

- Specifically, Blanchard and Quah considered a bivariate system $\mathbf{y}_t = (q_t, ur_t)'$, where q_t is output growth and ur_t is unemployment rate. They assumed that the respective structural innovations represent supply and demand shocks.

- Blanchard and Quah identified the structural model by further assuming that the demand shocks only have transitory effects on output growth. That is, the sum of the impulse responses for all future periods from the demand shocks on output growth is zero.

- Formally, consider the MA representation of the SVAR model (23) in terms of the structural innovations, $\mathbf{u}_t$, and let $\mathbf{\Theta}_i$ denote the MA coefficient matrices. Identification can be achieved by placing zero restrictions on the matrix of total accumulated (or long-run) effects:

$$\mathbf{\Theta}_\infty \equiv \sum_{i=0}^{\infty} \mathbf{\Theta}_i = \left(\mathbf{I}_k - \mathbf{\Phi}_1 - \ldots - \mathbf{\Phi}_p\right)^{-1} \mathbf{A}^{-1}\mathbf{B}.$$

- In Blanchard and Quah's bivariate example, the (1,2)th element of the above matrix is restricted to zero. More generally, we could restrict the upper right-hand corner elements of $\mathbf{\Theta}_\infty$ to zero, which (given the VAR parameters) imposes $k\left(k-1\right)/2$ restrictions on $\mathbf{A}^{-1}\mathbf{B}$. If we set $\mathbf{A} = \mathbf{I}_k$, the $\mathbf{B}$-model is just identified.

- The lower-triangular form of $\mathbf{\Theta}_\infty$ implies that u_{jt} has no long-run effect on $y_{\ell t}$ in the preceding equations ($\ell < j$).